
Recession Indicators

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Background

What Is a Recession?

A recession is a significant decline in economic activity across the economy, lasting more two or more quarters, typically visible in GDP, employment, income, and industrial production.

The Role of Forecasting:

- Helps businesses and policymakers prepare
- Reduces financial and employment risk
- Improves decision-making under uncertainty

Historical recession indicators:

- Yield Curve Inversion
- Rising Unemployment
- Slowing GDP
- Stock Market Declines
- Weak Consumer Spending
- Falling Industrial Activity

Background cont.

What Has Amplified Previous U.S. Recessions:

- 2020: COVID-19 shutdowns, supply-chain breaks, collapse in spending
- 2008: Housing crash, bank failures, frozen credit markets
- 2001: Dot-com bubble burst, 9/11 shock
- 1990: Oil price spike, drop in consumer confidence
- 1981: High interest rates, Fed tightening, credit contraction

Potential Future Risk Factors:

- Debt
- Fragile supply chains
- Fast rate hikes
- Global tensions
- Financial volatility
- Tech/AI bubbles.

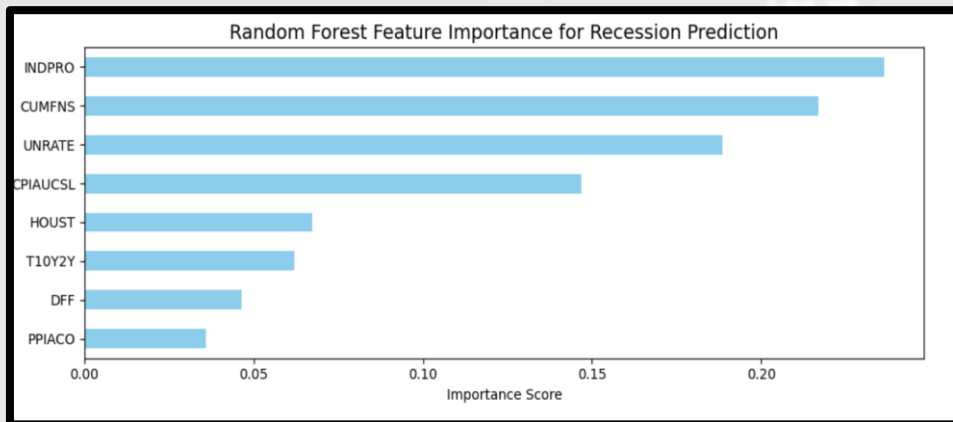
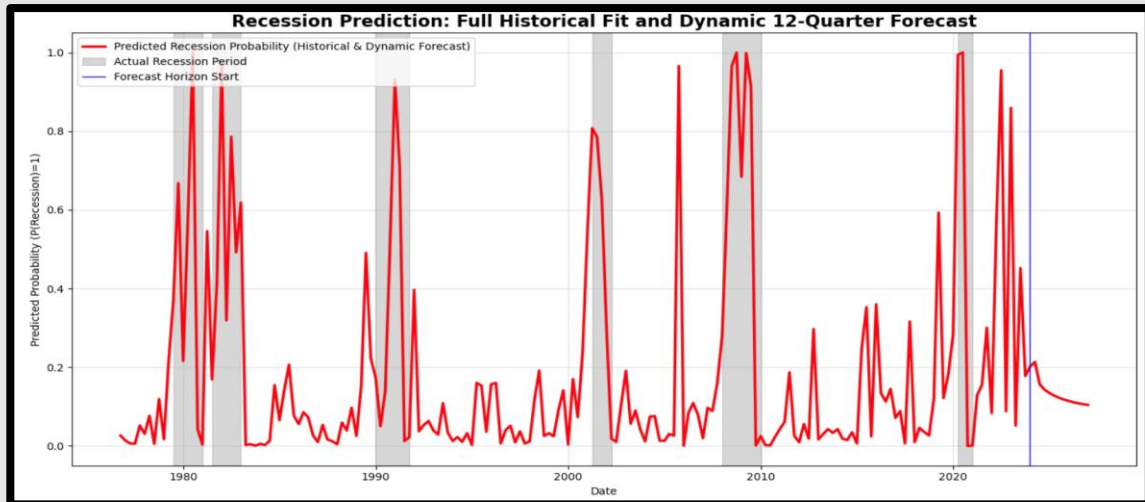
Implications:

- Recession amplifiers show underlying weaknesses met by unexpected shocks can lead to recessions.
- By relying on key macroeconomic indicators to identify early signs, they can assist to forecast potential recessions before shocks intensify the downturn.

Question?

**How accurately can
macroeconomic
data forecast U.S.
recessions?**

Model 1 – Random Forest -> PCA -> Logistic Regression



--- Model Evaluation (Out-of-Sample) ---
Accuracy Score: 0.8684
Classification Report (Focus on Class 1 - Recession):

	precision	recall	f1-score	support
0.0	1.00	0.86	0.93	36
1.0	0.29	1.00	0.44	2
accuracy			0.87	38
macro avg	0.64	0.93	0.68	38
weighted avg	0.96	0.87	0.90	38

--- Null Model Evaluation (Always Predicting 'No Recession') ---
Null Model Accuracy Score: 0.9565
Classification Report (Null Model):

	precision	recall	f1-score	support
0	0.96	1.00	0.98	44
1	0.00	0.00	0.00	2
accuracy			0.96	46
macro avg	0.48	0.50	0.49	46
weighted avg	0.91	0.96	0.94	46

88% of
months
since 1980
have not
had
recessions

Component 1: General Economic Health, Component 2: Inflationary Pressure

Model 2 Overview – Technical Recession Probability

Technical Recession (TechRec)

A recession defined as two straight quarters of negative Gross Domestic Product (GDP) growth.

Target variable:

1 = recession occurs

0 = no recession

Goal: detect if the economy is entering a downturn or close to one.

Model Inputs:

- GDP growth
- Industrial production growth
- Unemployment level
- Change in unemployment
- Yield curve spread (10-year minus 3-month interest rate)
- Consumer spending growth
- Stock market return

Model 2 Performance – (Technical Recession Probability)

Metric	Value	Meaning
Accuracy	0.86	Model predicts “no recession” correctly most of the time.
Precision	0.00	The model never predicted a recession in the test set, because none occurred.
Recall	0.00	No recessions occurred in the test data → nothing to detect.
Brier Score	0.021	Very low error in probability estimates.
ROC AUC	Not computable	Only one class (no recession) existed in the test set.

Interpretation:

The United States did not experience a technical recession in the test period, so the model correctly predicted “no recession” every quarter.

Model 3 Overview – NBER Recession Start Probability

NBER Recession Start:

- Official recession designation by the National Bureau of Economic Research.
- Triggered when economic activity (employment, income, spending, production) declines broadly.

Target variable:

1 = recession begins within the next 12 months

0 = no recession

Goal: estimate risk of an official recession announcement.

Model Inputs:

- Yield curve spread (10-year minus 3-month interest rate)
- Two-quarter moving average of the yield curve spread
- Unemployment level
- Change in unemployment
- Industrial production growth
- Consumer spending growth
- Stock market return

Model 3 Performance – (NBER Recession Start Probability)

Metric	Value	Meaning
Accuracy	0.38	Several false alarms because model predicted slight risk.
Precision	0.00	Model predicted small risks, but no recessions occurred.
Recall	0.00	No true recessions to detect.
Brier Score	0.115	Error is still low for rare-event modeling.
ROC AUC	Not computable	Only one class occurred in the test data.

Explanation:

There was no recession in the test period, so the model had no true events to detect.

Scenario Probabilities – (Using Today's Economic Values)

Model 2: Technical Recession Probability

Scenario	Probability	Interpretation
Baseline	1.1%	Current data shows almost no recession risk.
Pessimistic shock	0.15%	Even with a shock, recession risk stays very low.
Optimistic	5.2%	Strong economy raises growth but still low recession risk.

Model 3: NBER Recession Start Probability

Scenario	Probability	Interpretation
Baseline	3.0%	Very low chance of an official recession starting soon.
Pessimistic shock	0.6%	Shock has small effect because model sees no downturn signals.
Optimistic	6.6%	Strong improvements raise activity, but recession risk stays low.

Both models agree:

Based on today's conditions, the U.S. economy is not near a recession under any realistic scenario.

Model 4 Overview – Vector Autoregression (VAR) Forecast

Vector Autoregression (VAR)

- A statistical model that uses past values of multiple economic indicators to forecast their future values together.

Purpose of this model:

- Predict how key macroeconomic variables will evolve over the next several quarters.
- Provide forecasted macro inputs for the logistic recession models.
- Allow recession probabilities to be computed for future periods instead of only today.

Why this matters:

- Recession risk depends on where the economy is headed.
- A VAR captures relationships among economic indicators and projects them forward.

Model Inputs:

- Gross Domestic Product (GDP) growth
- Industrial production growth
- Change in unemployment
- Yield curve spread (10-year minus 3-month interest rate)

No target variable:

- This VAR has no target variable because it forecasts economic indicators, not recession outcomes.

Model 4 Context –

Key VAR Model Statistics

Statistic	Value	Simple Meaning
No. of Equations	4	Four macroeconomic variables were modeled together.
Nobs	33	Number of quarterly observations used to estimate the model.
Log likelihood	-49.2264	Indicates overall fit; higher is better for VAR.
AIC	-4.24688	Lower values mean better model fit relative to complexity.
BIC	-1.16316	Penalizes large models more strongly; confirms acceptable fit.
HQIC	-3.20930	Another measure of model quality; lower is better.
FPE	0.0221404	Forecast Prediction Error — lower values mean better accuracy.
Det(Omega_mle)	0.00420108	Measures residual variance; small values indicate stable forecasts.

How the VAR Supports Our Recession Models:

- The VAR forecasts future economic conditions, not recessions.
- These forecasts are fed into our logistic models to compute future recession probabilities.

Model 4 Results – Future Recession Probabilities (VAR forecasts)

Future Quarter	Probability	Interpretation
2026 Q2	7.4%	Small recession risk, not close to danger levels.
2026 Q3	5.0%	Risk declines as economy stabilizes.
2026 Q4	2.7%	Still low recession threat.
2027 Q1	2.2%	Continues trending downward.
2027 Q2	1.5%	Very low chance.
2027 Q3	1.4%	Economy appears to stay stable.

Key takeaway

Model 2: Technical Recession Probability (Using VAR Forecasts)

- The highest predicted risk is 7.4%, which is far below the 30% concern threshold.
- The model suggests no technical recession ahead.

Future Quarter	Probability	Interpretation
2025 Q4	12.6%	Small concern, but still low overall.
2026 Q1	0.3%	Almost no risk at this point.
2026 Q2	15.8%	Slight increase in recession signals.
2026 Q3	18.8%	Peak risk, still below danger level.
2026 Q4	12.2%	Risk begins to fall.
2027 Q1	13.2%	Moderate but not threatening.
2027 Q2	6.4%	Very low risk.
2027 Q3	4.8%	No signs of a recession forming.

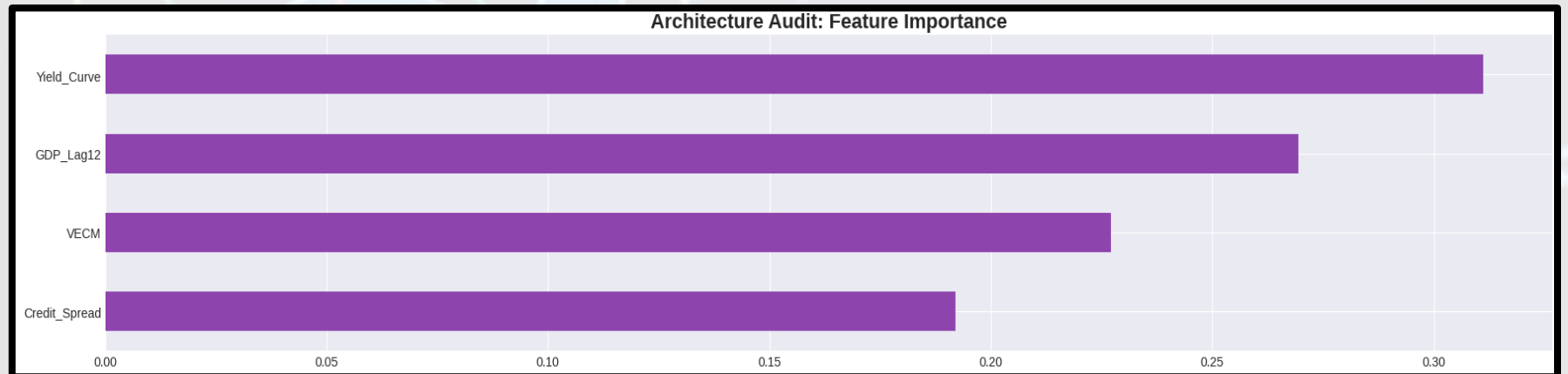
Key takeaway

Model 3: NBER Recession Start Probability (Using VAR Forecasts)

- The highest predicted probability is 18.8%, still well below the 30% threshold.
- Model forecasts no official NBER recession beginning in 2025–2027.

Model Comparison –

- Recession probability only spikes before real downturns.
- Recent signals stay well below the trigger threshold.



Summary –

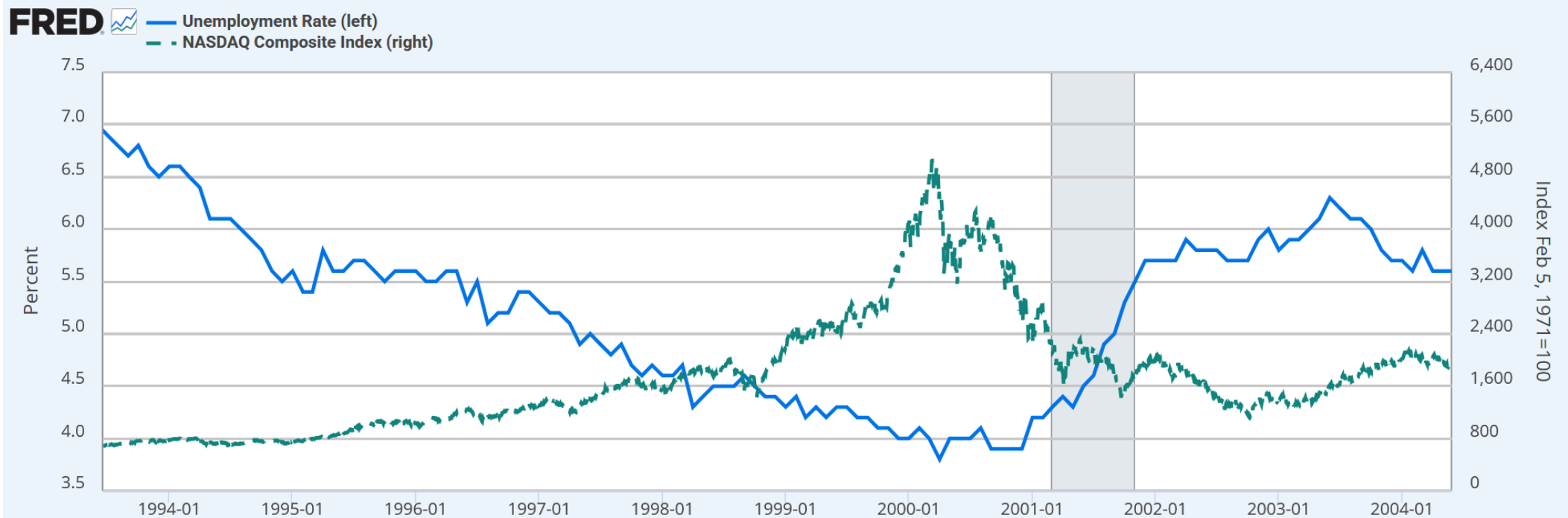
- Multiple models agree reflecting that current recession risk is low
- Key indicators show stable economic conditions moving forward
- VAR forecasts support continued resilience, not downturn
- A multi model approach provides more reliable signals than any single metric



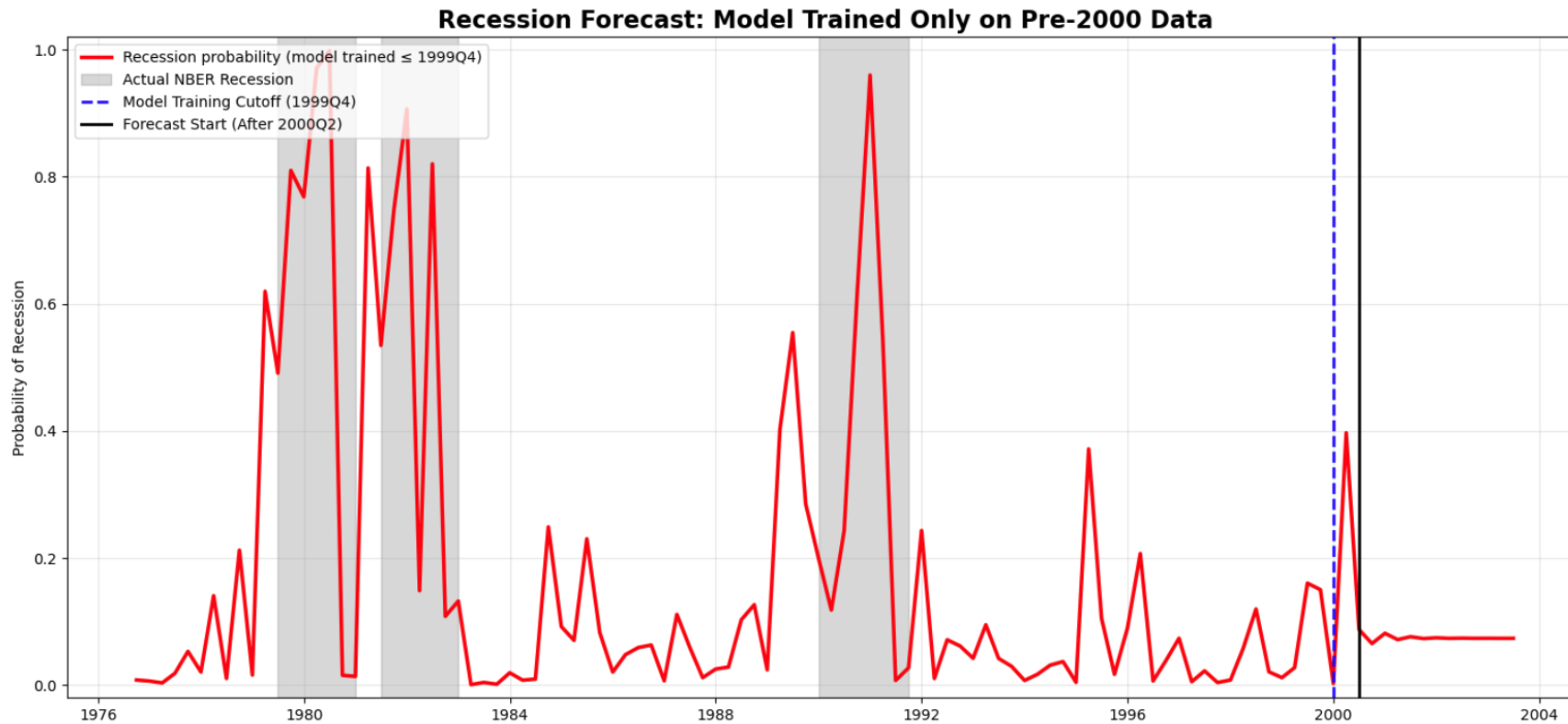
Historic Consideration: The 2001 Dot-com Bubble



- In the 1990s, image-based internet access completely changed commerce
- Valuations of internet companies skyrocketed unsustainably until the bubble burst



Economics can be misleading



Prompt: "Is your parent company losing money with every AI prompt?"

ChatGPT: "Recent projections suggest that for every dollar of revenue, OpenAI is spending \$1.69"

Gemini: "For a single interaction, the cost of running the free version of Gemini is likely higher than the revenue it directly generates"

Grok: "xAI is reportedly burning through about \$1 billion per month. Analysts project it could become profitable by 2027 or 2026"

Thank You

Questions?